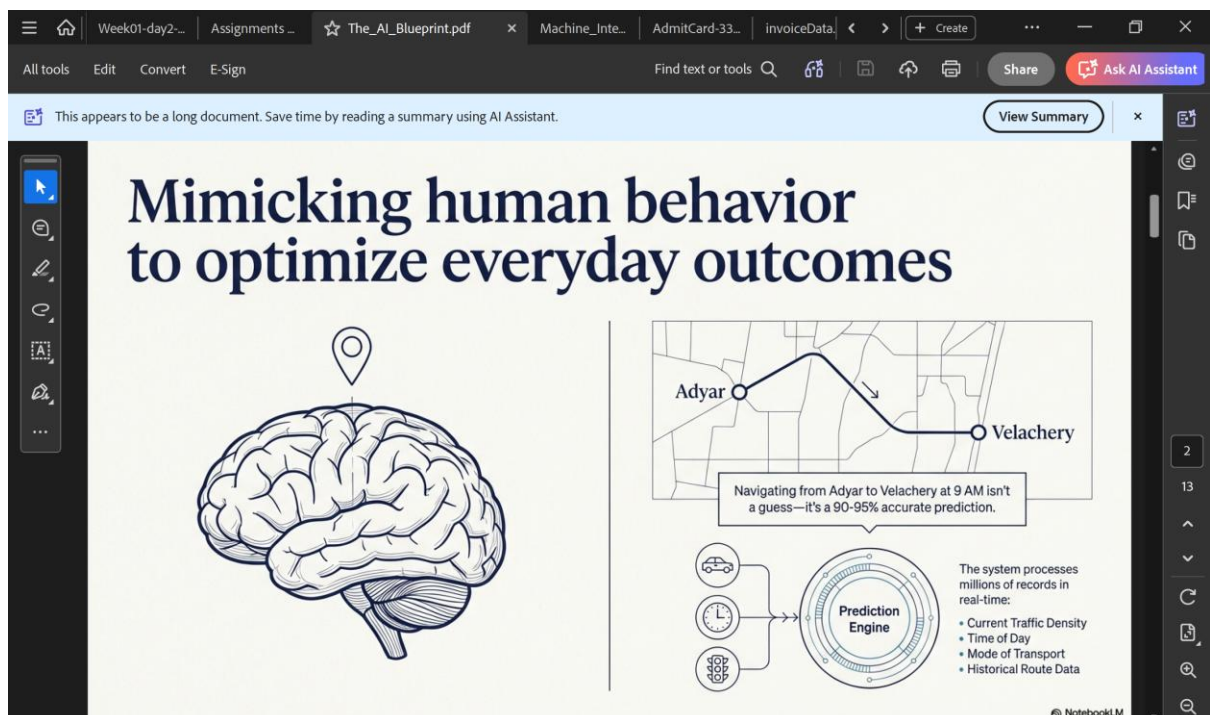
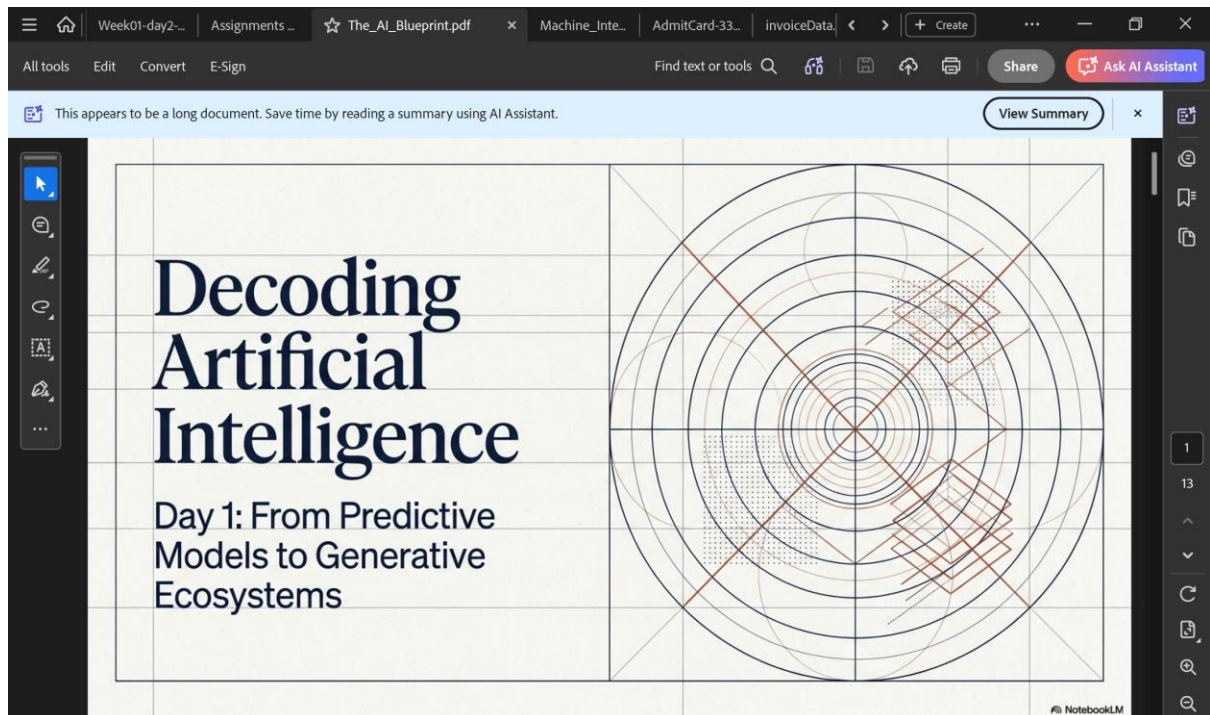


1) Use notebooklm to convert Classroom file Agenda--Day1 to ppt or deck

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Layer 1: Machine Learning (ML)

Core Concept: A predictive subset of AI.
Scale of Operation: Highly efficient on 1,000s to Millions of records.
Primary Output: Prediction.

1,000s Millions

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How the Model Learns

```
graph LR; S1[Step 1: Data Collection] --> S2[Step 2: Data Cleansing]; S2 --> S3[Step 3: Model Selection  
(e.g., Convolutional Neural Networks)]; S3 --> S4[Step 4: Train & Test]; S4 --> S4_1[80% Train]; S4 --> S4_2[20% Test & Validate];
```

The 80/20 Split

- 80% of data is used to Train the model.
- 20% of data is reserved to Test & Validate the model.
- Testing relies on input data the model has never seen before.

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Categorizing the Data Structure



Supervised Learning

- **Mechanism:** Uses explicitly Labelled data (e.g., Content star ratings, likes/dislikes).
- **Linear Regression:** Delivers numerical outputs (e.g., Real estate pricing, defect prediction).
- **Classification:** Delivers categorical labels (e.g., Google predicting heart attack risk: Yes or No).



Unsupervised Learning

- **Mechanism:** Uses Unlabelled data.
- **Focus:** Identifies hidden behaviors, genres, and patterns.
- **Example:** Collaborative filtering that groups users by behavior without explicit likes or comments.

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Predictive Machine Learning in Action

1. The Parameters
(Millions of Records)

Age	Blood Group Diabetic	Blood Pressure	Location Phone	Genetic History Smoker	Smoker Drinker
•	•	•	•	•	•
•	•	•	•	•	•
•	•	•	•	•	•
•	•	•	•	•	•
•	•	•	•	•	•
•	•	•	•	•	•

2. The Unseen Input



3. The Classification Output

CNN ModelHeart Disease Risk: YES

A net-new Retinal Image is fed into the Convolutional Neural Network (CNN).

(Note: Capable of multimodal ingestion including text, image, or video).

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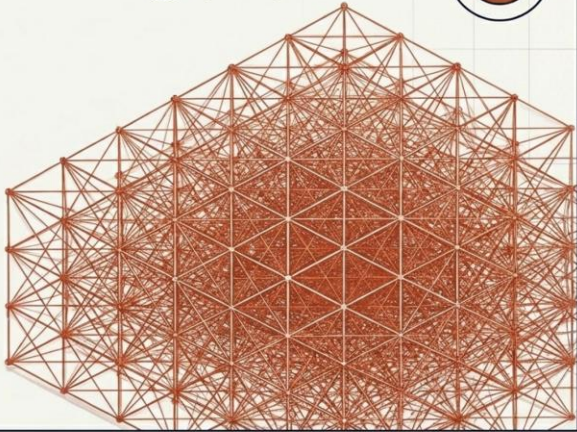
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Layer 2: Deep Learning (DL)

Core Concept: Machine Learning infused with Neural Networks.

Scale of Operation: Scales dramatically from Millions to Trillions of records.

Primary Output: Advanced Predictive Modeling across vast, unstructured multimodal data.



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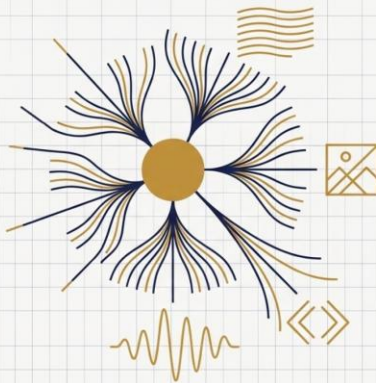
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Layer 3: Generative AI

Core Concept: A subset of Deep Learning that shifts the paradigm from Predicting to Generating.

The Output:
Instead of numbers or categories, Gen AI creates net-new:

- Text
- Images
- Audio
- Video
- Code (applicable across all domains)



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Inside Large Language Models (LLMs)

The Engine: Generative Pretrained Transformers (GPT)

- Service Provider: OpenAI
- Chat Agent: ChatGPT
- Pretrained on vast, cross-domain datasets far beyond software alone.

The Architecture of Scale:

GPT-4:
96 to 120 Layers

GPT-5:
190 Layers | 120 Billion Parameters



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The AI Spectrum at a Glance

Machine Learning (ML)	Deep Learning (DL)	Generative AI (Gen AI)
- Focus: Predictive - Data Scale: 1,000s to Millions of records - Output: Numbers (Regression) & Categories (Classification)	- Focus: Advanced Predictive - Data Scale: Millions to Trillions of records - Output: Multimodal diagnostics (Images, Text, Video)	- Focus: Generative - Data Scale: Billions of Parameters - Output: Net-new Original Content & Code

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Next Up: Day 2

Deep dive into LLM anatomy and internal working mechanics.

The diagram illustrates the internal structure of a Large Language Model (LLM) as a stack of layers. From bottom to top, the layers are: Input Embedding, Multi-Head Attention, Add & Norm, Feed Forward Network, Add & Norm, and Softmax. Arrows indicate the flow of data through these layers, showing how the input is processed through multiple attention heads and feed-forward networks, with residual connections (Add & Norm) being added at each stage. The final output is passed through a Softmax layer for token prediction.

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